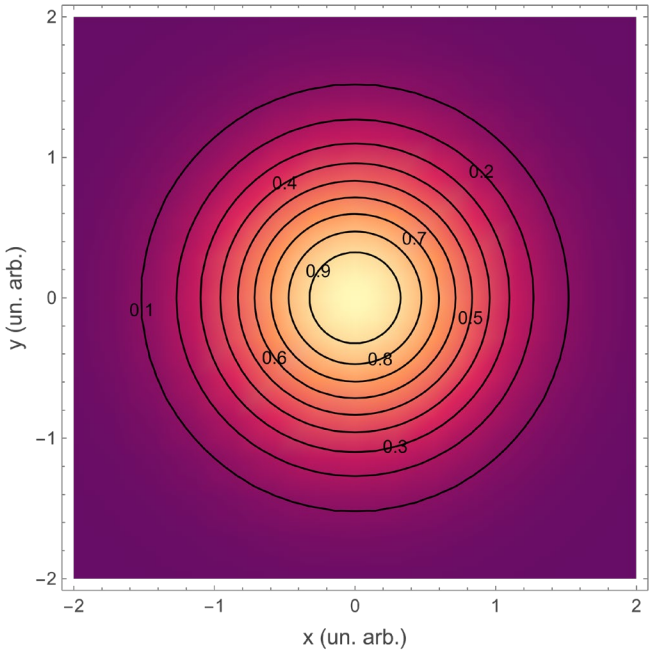
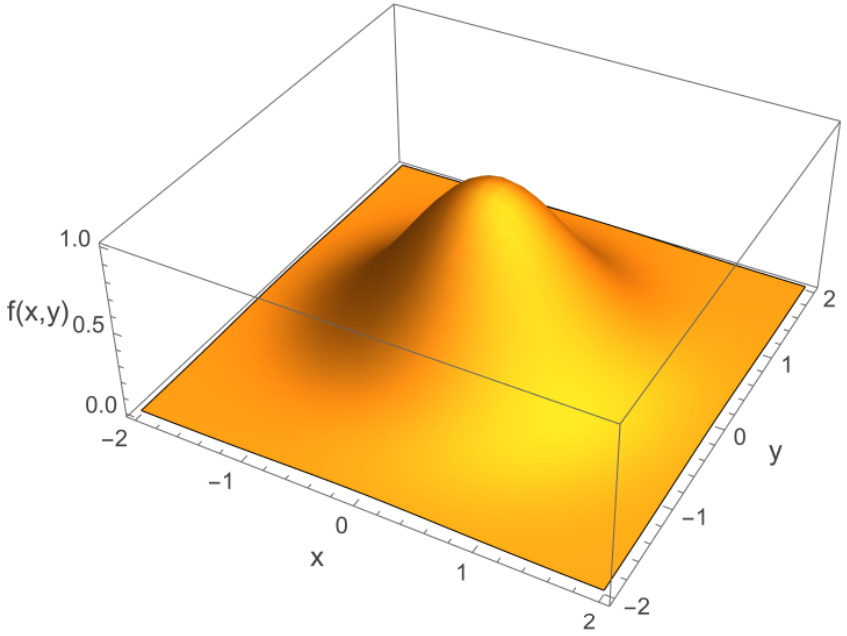
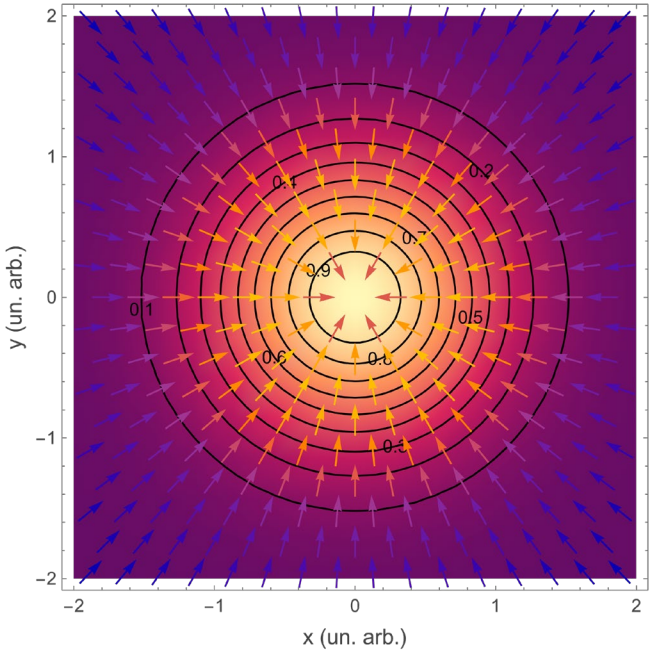
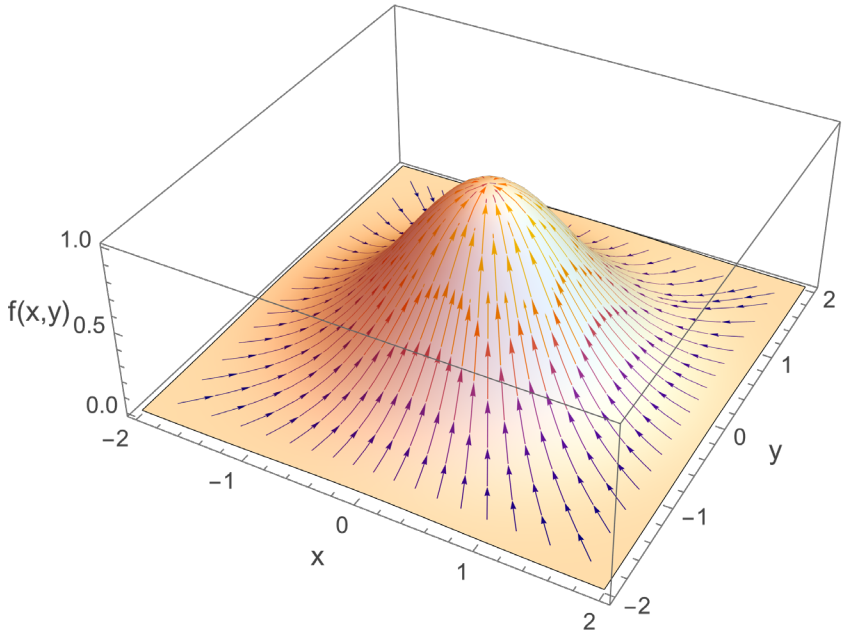
$$f\ (x,y) = e^{-(x^2+y^2)}$$

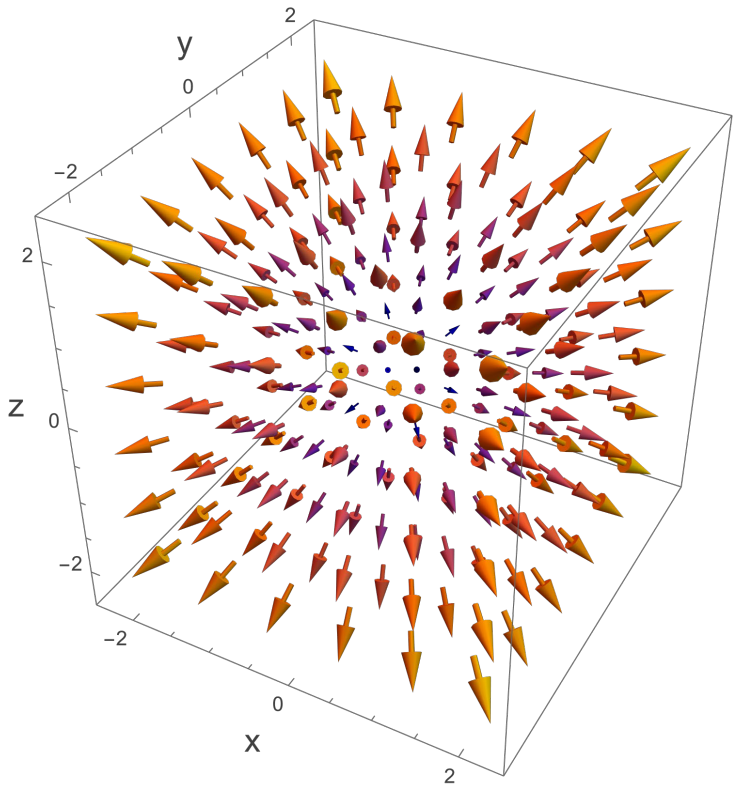


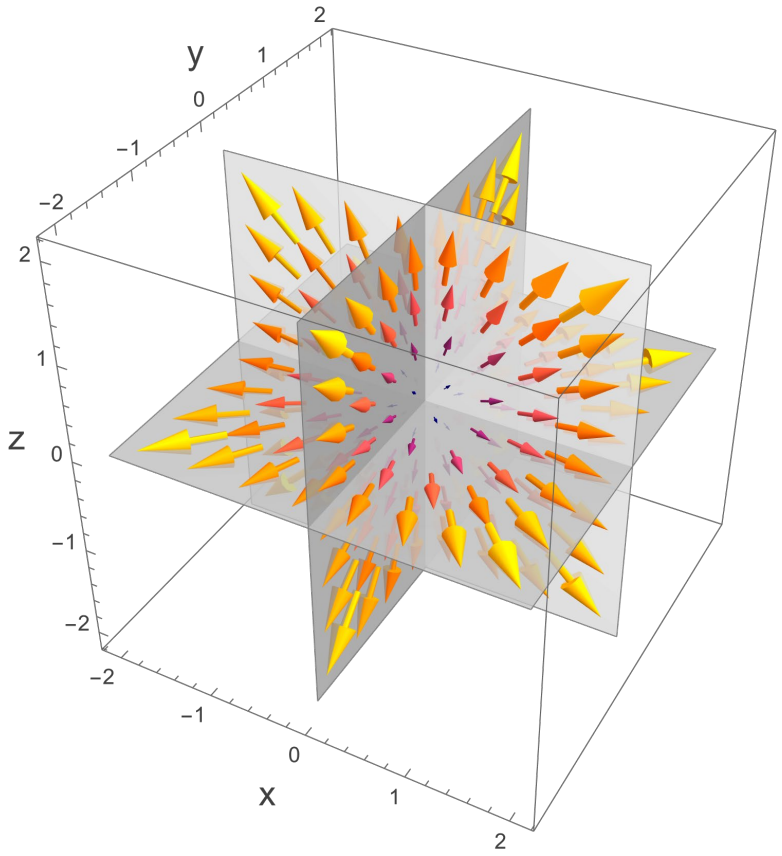


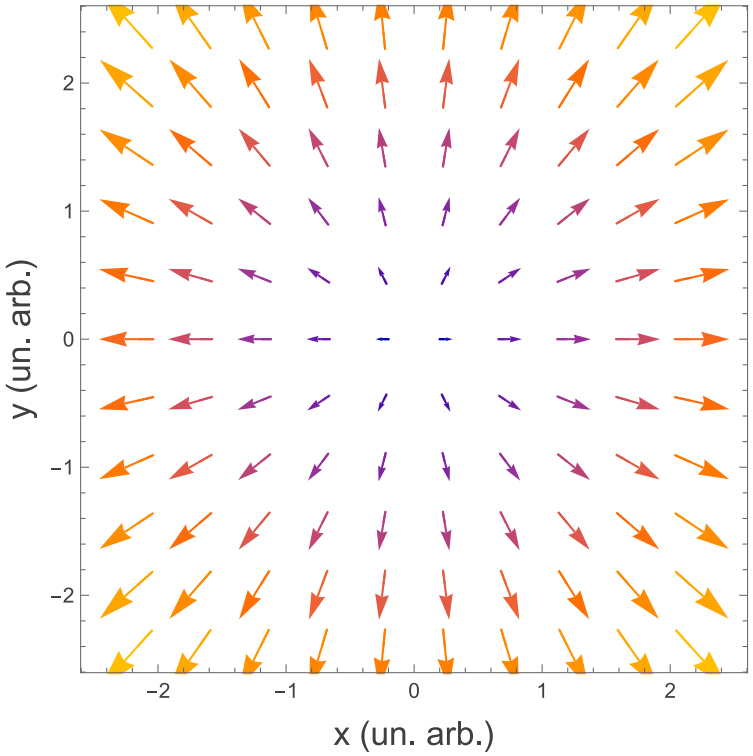
$$\vec{\nabla} f = -2e^{-(x^2+y^2)}\left(x\ \hat{u}_x + y\ \hat{u}_y\right)$$



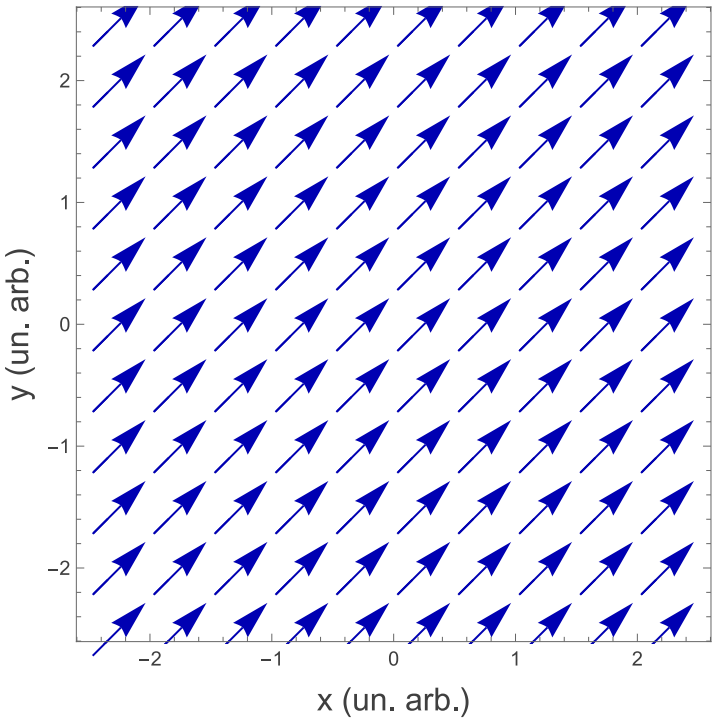




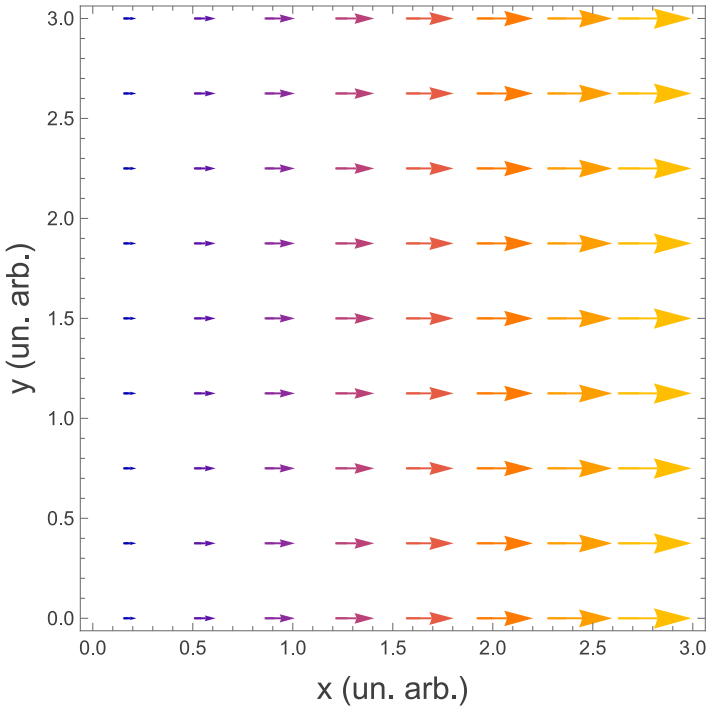




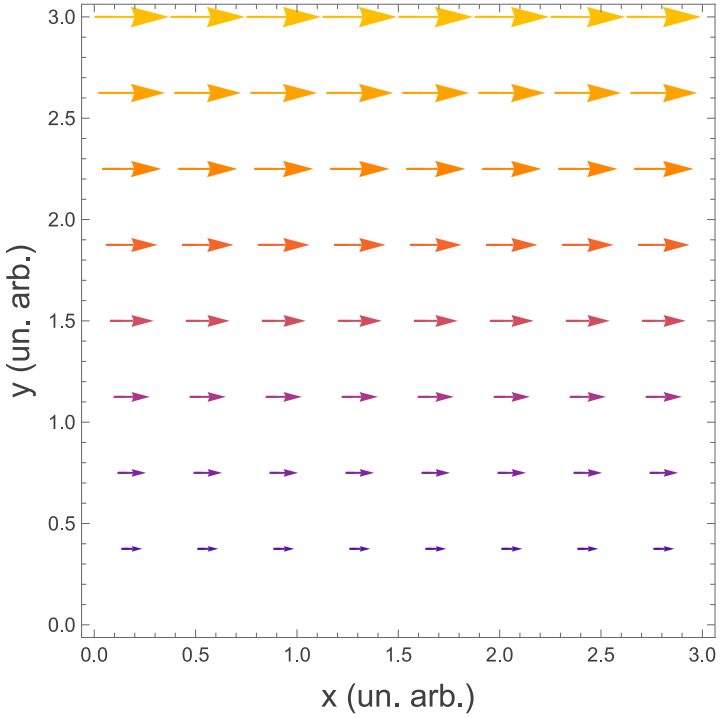
$$\vec{F}\,(\vec{r}) = x\,\hat{u}_x + y\,\hat{u}_y + z\,\hat{u}_z$$



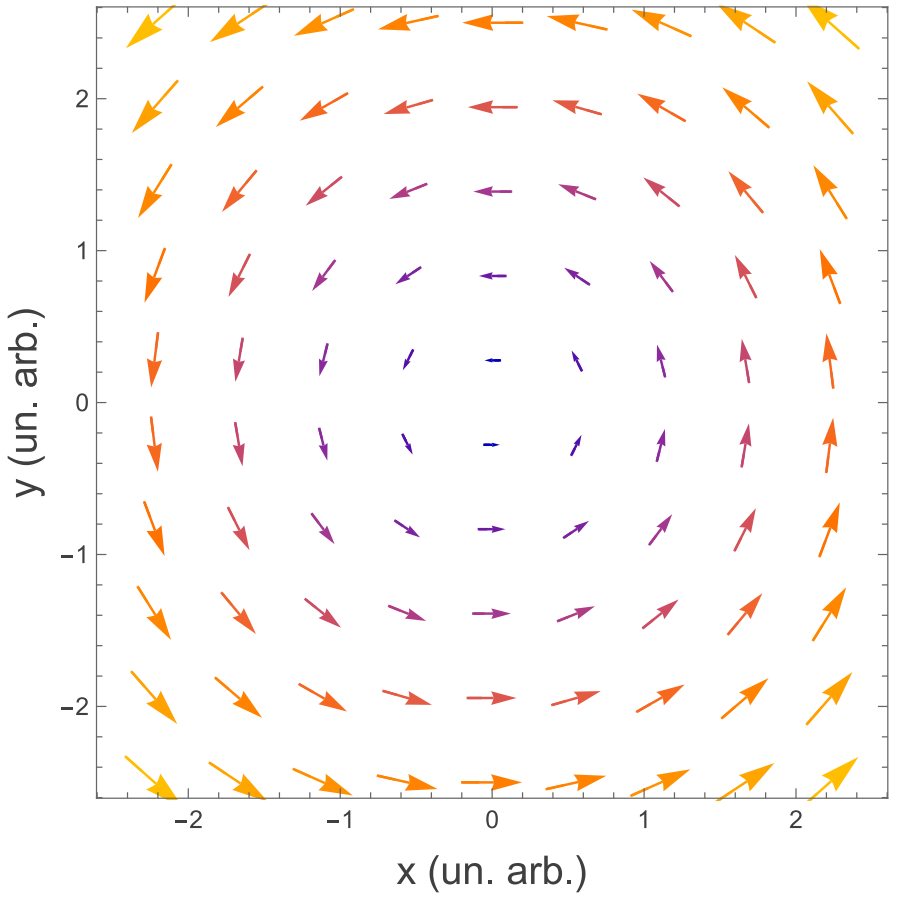
$$\vec{F}\,(\vec{r}) = \hat{u}_x + \hat{u}_y$$



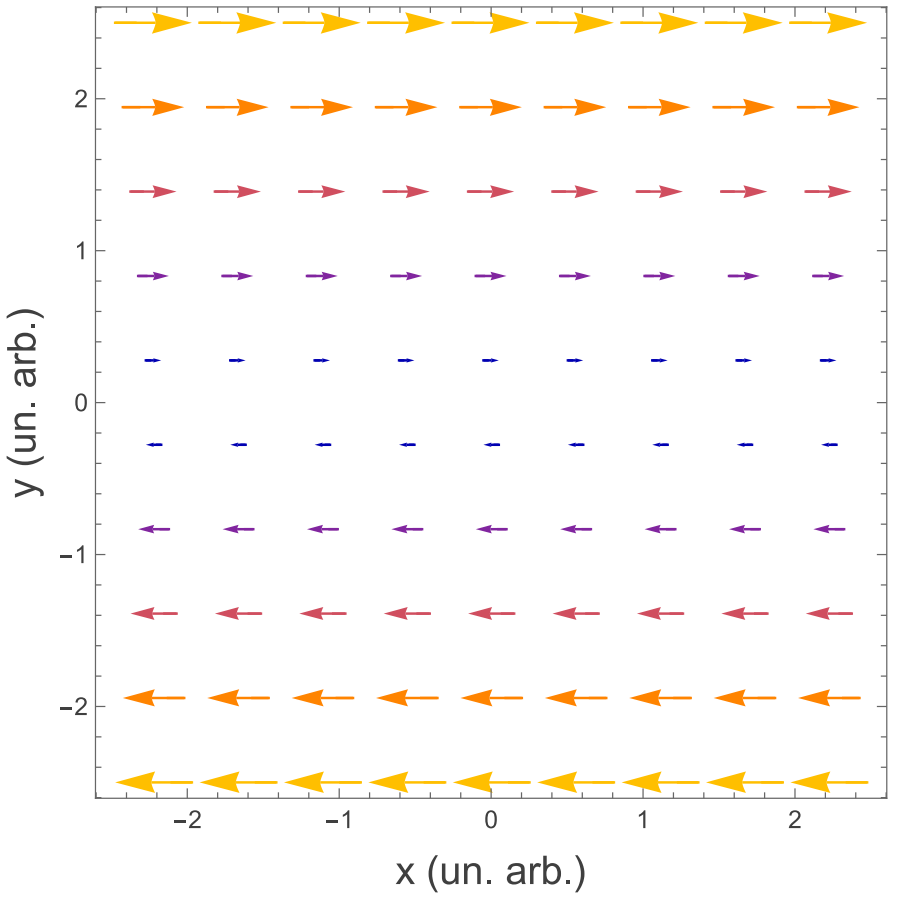
$$\vec{F}\,(\vec{r}) = x\,\hat{u}_x$$



$$\vec{F}\,(\vec{r}) = y\,\hat{u}_x$$

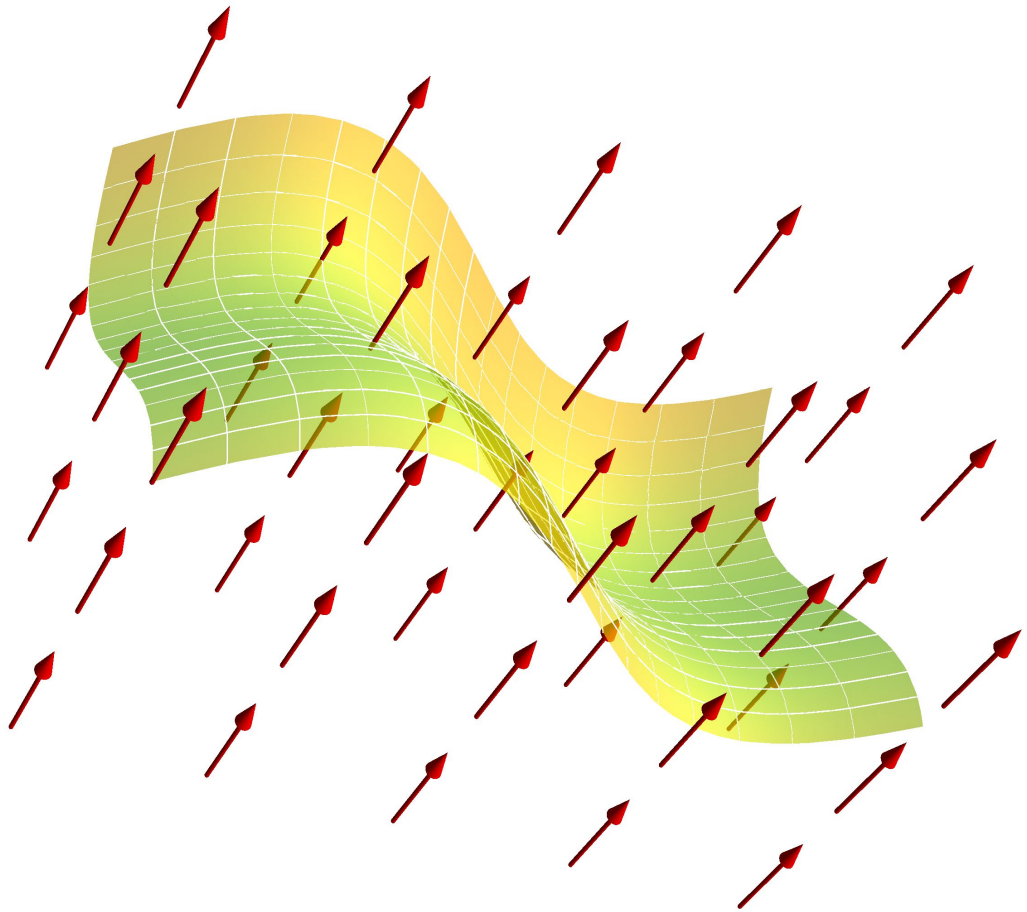


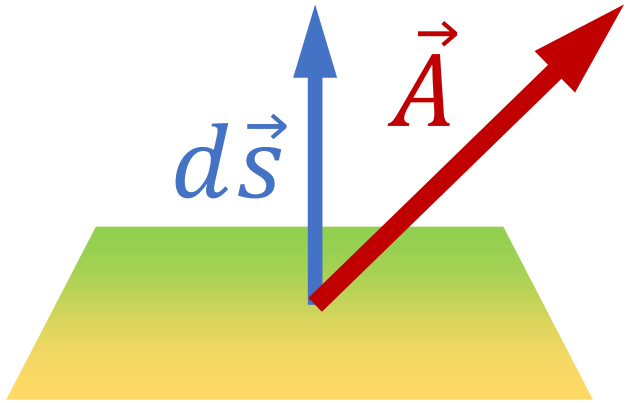
$$\vec{F}\,(\vec{r}) = -y\,\hat{u}_x + x\,\hat{u}_y$$

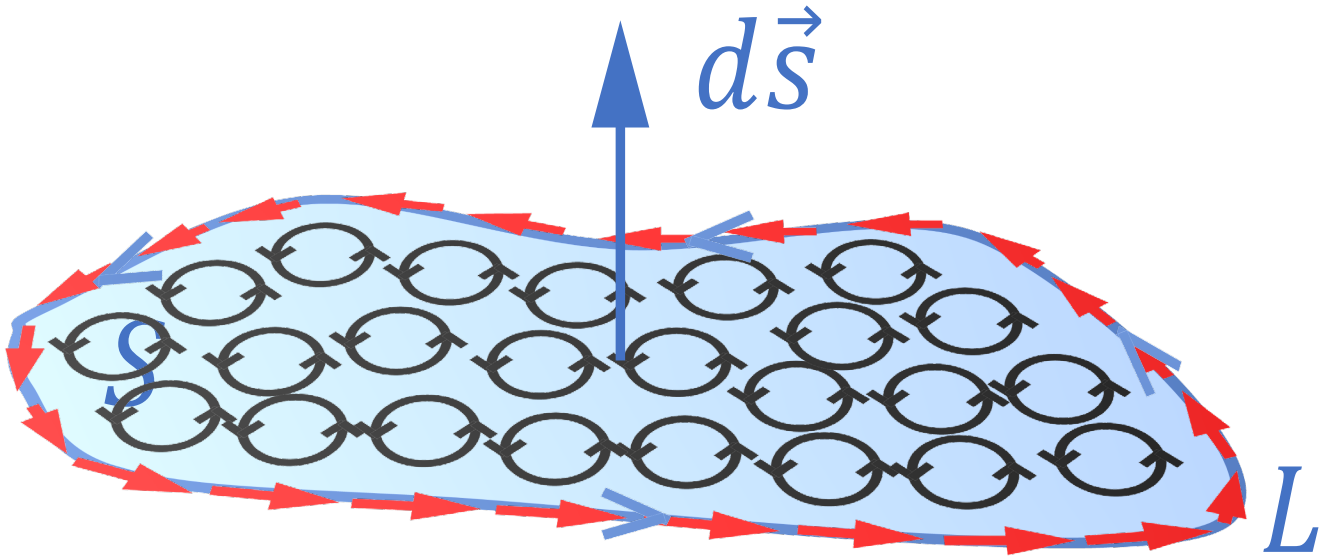


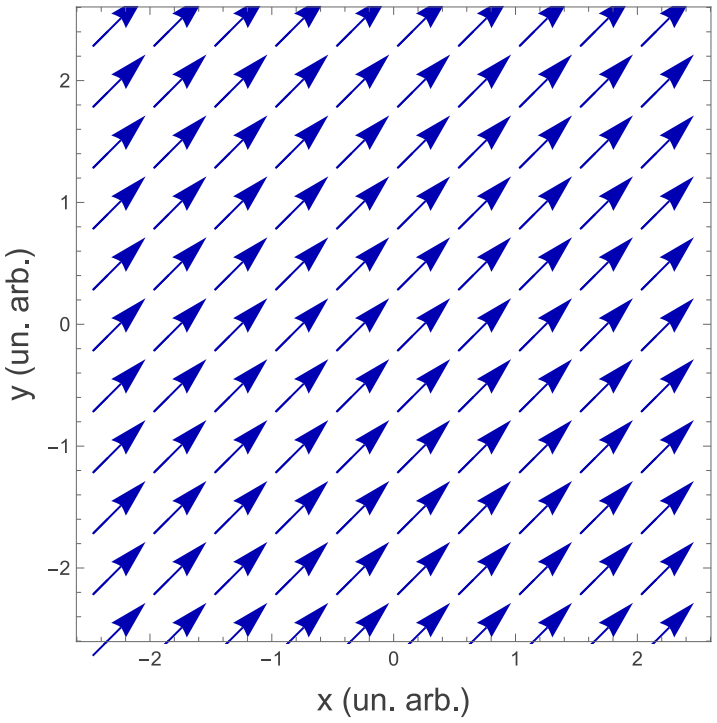
$$\vec{F}\,(\vec{r}) = y\,\hat{u}_x$$

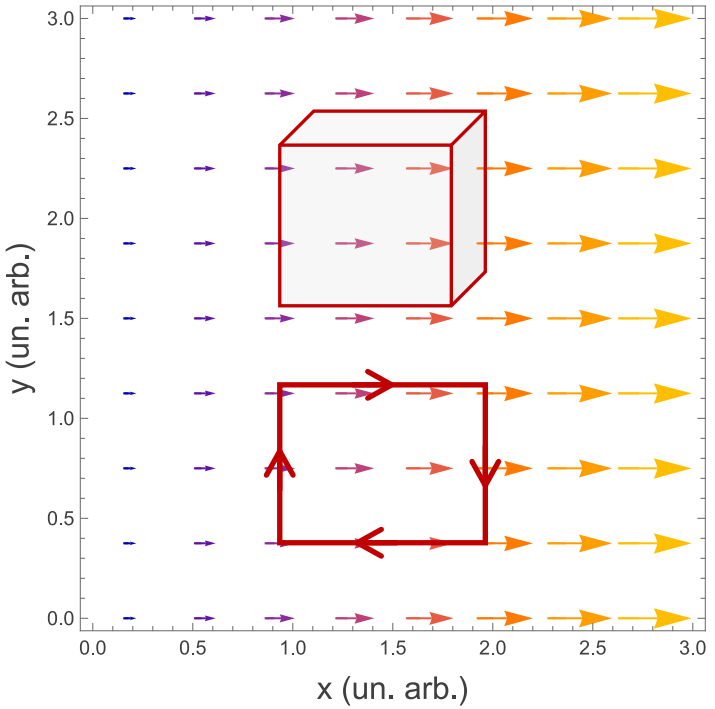
https://www.youtube.com/watch?v=kJYoV8Kq6Vo

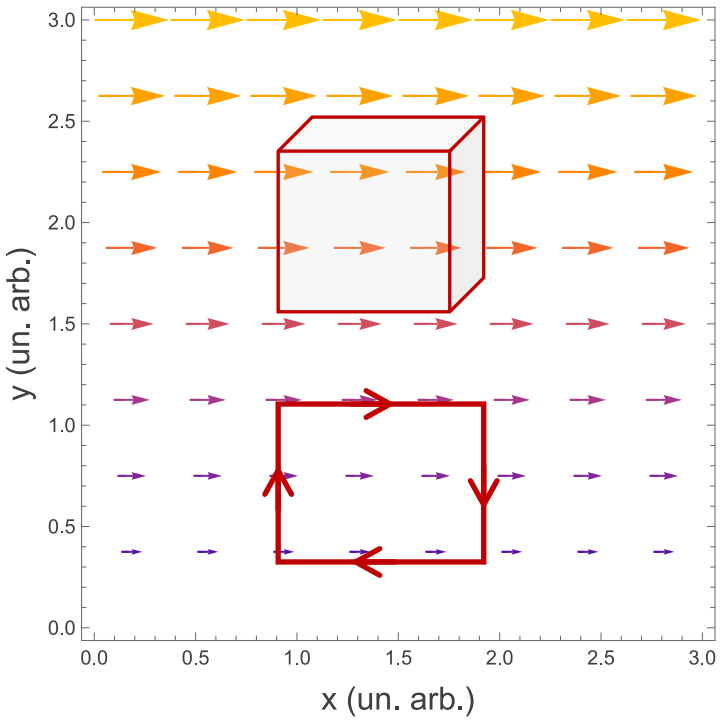



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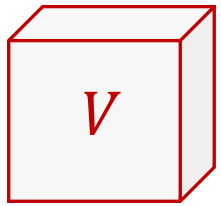


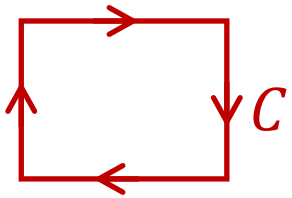


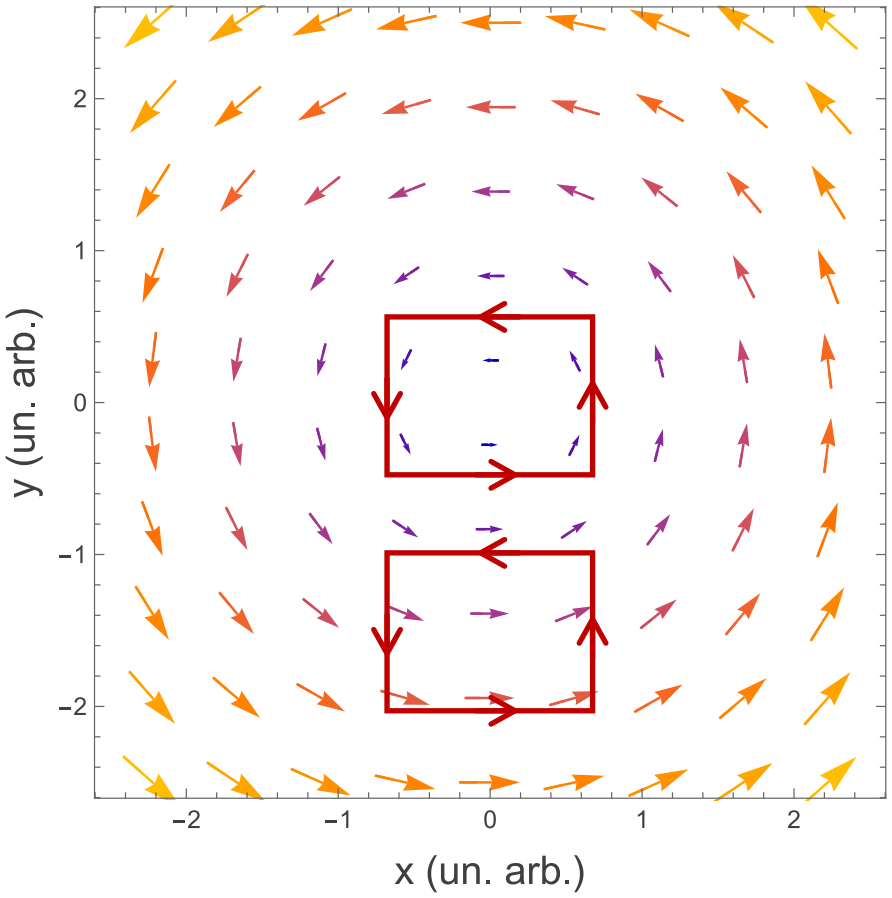
$$\vec{F}\,(\vec{r}) = \hat{u}_x + \hat{u}_y$$

$$\vec{F}\,(\vec{r}) = x\,\hat{u}_x$$

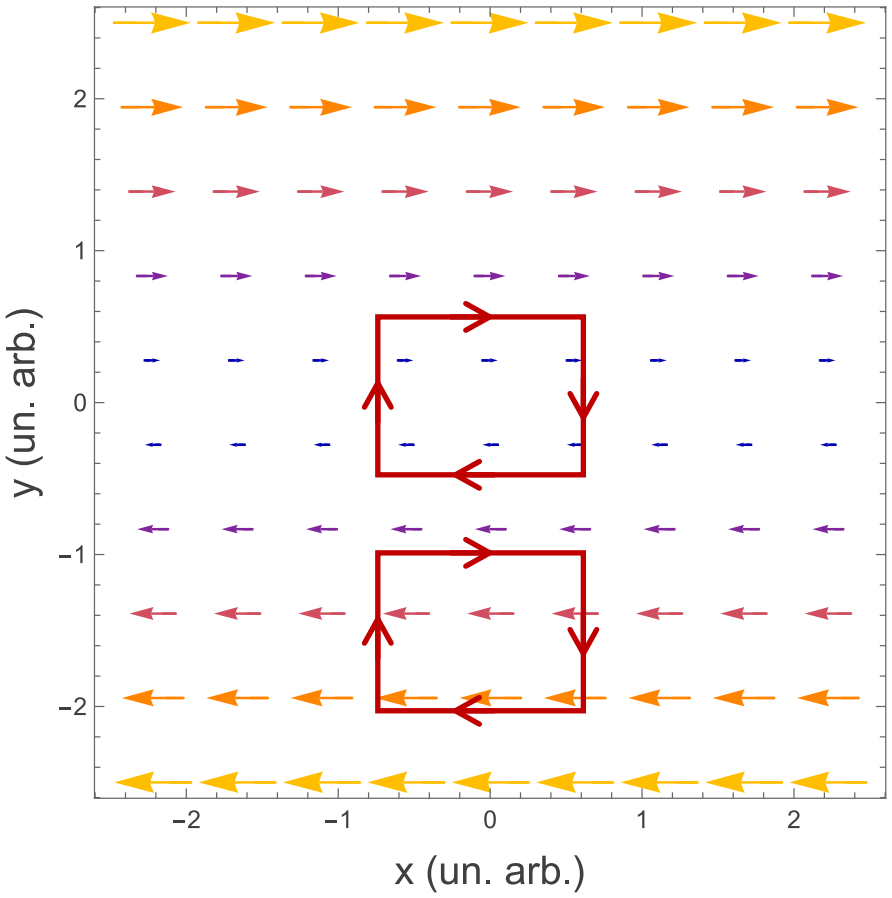
$$\vec{F}\,(\vec{r}) = y\,\hat{u}_x$$







$$\vec{F}\,(\vec{r}) = y\,\hat{u}_x - x\,\hat{u}_y$$



$$\vec{F}\,(\vec{r}) = y\,\hat{u}_x$$

